INCH-POUND

MIL-DTL-32628/1
11 April 2019

DETAIL SPECIFICATION SHEET

BACKSHELL, ELECTRICAL CONNECTOR, TRI-DENT SPRING BAND

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32628.

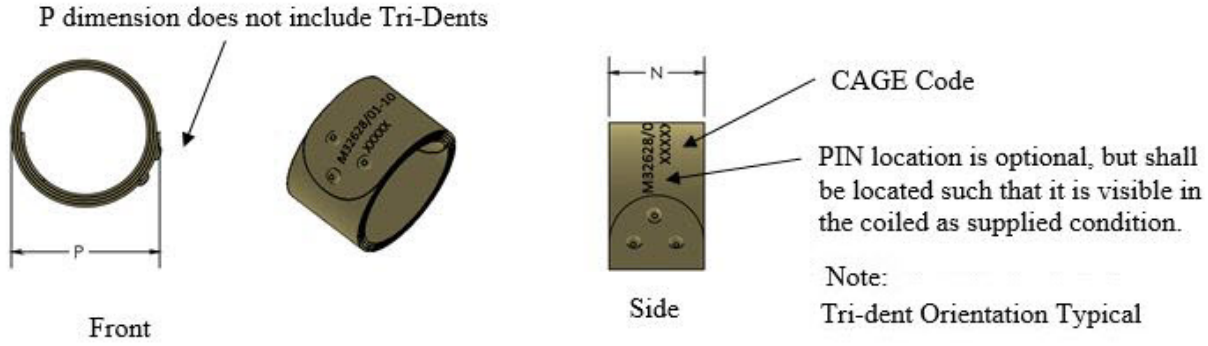


FIGURE 1. Tri-Dent splice configuration (M32628/01-10).

TABLE I. Spring band part number, material, and size.

M32628/01-XX Part number	Material	Size	
		P Diameter 1/ (±.030 inch)	N Width 1/ (±.030 inch)
-10	Stainless Steel in accordance with ASTM A666	.35	.31
-11		.63	
-12		.92	
-13		1.09	
-20	Non-magnetic cobalt alloy in accordance with SAE AMS5876	.35	
-21		.63	
-22		.92	
-23		1.09	

1/ See figure 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:15Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

1. Dimension are in inches.

2. The spring band outline configuration is optional within the dimensions shown (see figure 1).

3. Weight. See table II.

TABLE II. Weight.

M32628/01-XX Part number	Max weight (1000 count) (lbs.)
-10, -20	2.87
-11, -21	6.97
-12, -22	16.91
-13, -23	19.84

4. Braid retention pull force. See table III.

TABLE III. Braid retention pull force.

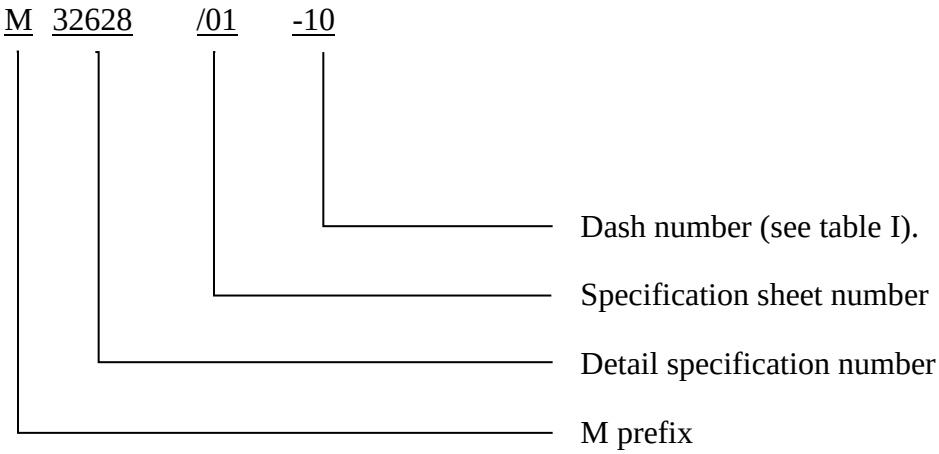
M32628/01 – XX Part number	A DIA (Maximum) 1/ (inches)	Braid retention pull force (lbs.)
-10, -20	.438	100
-11, -21	.625	150
-12, -22	.812	
-13, -23	1.125	

1/ See figure 3 and table VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:15Z
Check the source to verify that this is the current version before use.

5. Part or Identification Number (PIN):

Example:



EXAMINATIONS AND TESTS:

1. First article inspection and conformance inspection testing shall consist of the examinations and tests of MIL-DTL-32628 with the following modifications.

2. First article test samples (see tables IV and V). M85049/82-10W-03, M85049/82-18W-03 and M85049/82-28W-03 backshells shall be used for first article testing of M32628/01-10, -12, -13 or M32628/01-20, -22 and -23 spring bands, respectively.

- For the backshells M85049/82-10W-03, use 34-0500 diameter shield braid (part number AA59569R34T0500);

- For backshells M85049/82-18W-03, use 30-1000 diameter shield braid (part number AA59569R30T1000);

- For backshells M85049/82-28W-03, use 30-1375 diameter shield braid (part number AA59569R30T1375).

- A total of three sample assemblies plus one spare backshell, shield braid and spring band of each size and material for M32628/01-10, -12, -13 (stainless steel) or M32628/01-20, -22 and -23 (non-magnetic cobalt) spring bands, depending on spring band material being procured. See table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:15Z
Check the source to verify that this is the current version before use.

TABLE IV. First article test assembly samples.

Sample	Backshell	Shield braid	Spring band
Braid retention			
1	M85049/82-10W-03	AA59569R34T0500	M32628/01-10
2	M85049/82-18W-03	AA59569R30T1000	M32628/01-12
3	M85049/82-28W-03	AA59569R30T1375	M32628/01-13
Thermal aging			
4	M85049/82-10W-03	AA59569R34T0500	M32628/01-10
5	M85049/82-18W-03	AA59569R30T1000	M32628/01-12
6	M85049/82-28W-03	AA59569R30T1375	M32628/01-13
Re-installation test			
7	M85049/82-10W-03	AA59569R34T0500	M32628/01-10
8	M85049/82-18W-03	AA59569R30T1000	M32628/01-12
9	M85049/82-28W-03	AA59569R30T1375	M32628/01-13

TABLE V. Test sample components (including 1 spare).

Quantity	Component
4	M85049/82-10W-03 Backshells
4	M85049/82-18W-03 Backshells
4	M85049/82-28W-03 Backshells
4	AA59569R34T0500 Shield braid (6 inches each)
4	AA59569R30T1000 Shield braid (6 inches each)
4	AA59569R30T1375 Shield braid (6 inches each)
4	M32628/01-10 or M32628/01-20 Spring band
4	M32628/01-12 or M32628/01-22 Spring band
4	M32628/01-13 or M32628/01-23 Spring band

3. Braid retention. Shield braid shall not pull out and spring band slippage shall not exceed .025 inch when measured from a fixed point. Braid breakage during testing shall not be considered as a failure. One complete assembly for each backshell size with applicable shield braids and spring bands (see table I) shall be tested for a total of three complete assemblies:

a. M85049/82-10W-03 backshell assembly assembled with shield braid part number AA59569R34T0500, and spring band M32628/01-10 or -20, pull the braid at a rate of one inch per minute to a force of 100 pounds (see table III).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:15Z
Check the source to verify that this is the current version before use.

b. M85049/82-18W-03 backshell assembly assembled with shield braid part number AA59569R30T1000, and spring band M32628/01-12 or -22, pull the braid at a rate of one inch per minute to a force of 150 pounds (see table III).

c. M85049/82-28W-03 backshell assembly assembled with shield braid part number AA59569R30T1375, and spring band M32628/01-13 or -23, pull the braid at a rate of one inch per minute to a force of 150 pounds (see table III).

NOTES:

1. Intended use. M32628/01 spring bands are intended to be used with M85049/82 through /90 connector backshells in accordance with SAE AS85049/82 through SAE AS85049/90 (see table VI). These spring bands are intended to be used as an alternate for the M85049/128 shield band in accordance with SAE AS85049/128. In addition, the M32628/01 spring bands include Tri-Dents (see figure 1) to aid in installation of the spring band on the connector backshell.

2. Backshell compatibility. Spring band part number compatible with the M85049/82 through /90 backshells and shield splice. Spring band part number compatible backshells are shown in table VI. Installation of the spring band on the backshells shown in table VI have been validated. Using the spring band on other backshells or accessories may not provide the specified performance requirements of MIL-DTL-32628 spring bands.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:15Z
Check the source to verify that this is the current version before use.

TABLE VI. M85049/82 through /90 backshell compatibility.

.

Spring band M32628/01-XX Material/size 1/	M85049 accessory and cable entry size 2/									
	/82 - XXXXX		/83 - XXXXX		/84 - XXXXX		/85 - XXXXX		/86 - XXXXX	
	P/N	A DIA (REF) 3/ (inches)	P/N	A DIA (REF) 3/ (inches)	P/N	A DIA (REF) 3/ (inches)	P/N	A DIA (REF) 3/ (inches)	P/N	A DIA (REF) 3/ (inches)
-10, -20	08W03	.270	08W03	.270	08W03	.270	08W03	.270	08W03	.270
-10, -20	10W03	.332	10W03	.332	10W03	.332	10W03	.332	10W03	.332
-10, -20	12W02	.332	12W02	.332	12W02	.332	12W02	.332	12W02	.332
-10, -20	12W03	.458	12W03	.458	12W03	.458	12W03	.458	12W03	.458
-10, -20	14W02	.458	14W02	.458	14W02	.458	14W02	.458	14W02	.458
-11, -21	14W03	.582	14W03	.582	14W03	.582	14W03	.582	14W03	.582
-11, -21	16W02	.520	16W02	.520	16W02	.520	16W02	.520	16W02	.520
-11, -21	16W03	.645	16W03	.645	16W03	.645	16W03	.645	16W03	.645
-11, -21	18W02	.645	18W02	.645	18W02	.645	18W02	.645	18W02	.645
-12, -22	18W03	.770	18W03	.770	18W03	.770	18W03	.770	18W03	.770
-11, -21	20W02	.645	20W02	.645	20W02	.645	20W02	.645	20W02	.645
-12, -22	20W03	.832	20W03	.832	20W03	.832	20W03	.832	20W03	.832
-12, -22	22W02	.704	22W02	.704	22W02	.704	22W02	.704	22W02	.704
-13, -23	22W03	.958	22W03	.958	22W03	.958	22W03	.958	22W03	.958
-12, -22	24W02	.770	24W02	.770	24W02	.770	24W02	.770	24W02	.770
-13, -23	24W03	1.020	24W03	1.020	24W03	1.020	24W03	1.020	24W03	1.020
-13, -23	28W02	.895	28W02	.895	28W02	.895	---	---	---	---
-13, -23	28W03	1.145	28W03	1.145	28W03	1.145	---	---	---	---
-12, -22	61W02	.770	---	---	---	---	---	---	---	---
-13, -23	61W03	1.020	---	---	---	---	---	---	---	---

1/ -1X = Stainless steel spring band; -2X = Non-magnetic cobalt alloy spring band.
2/ See figure 2 – last five characters of the part number.
3/ See figure 3. Note that in addition to the A dimension DIA, the thickness of the braid shield must be considered. Shield braid thickness was accounted for in determining appropriate spring band size.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:15Z
Check the source to verify that this is the current version before use.

TABLE VI. M85049/82 through /90 backshell compatibility – Continued.

Spring band M32628/01 - XX Material/size 1/	M85049 part number and shield termination area diameter 2/							
	/87 - XXXXX		/88 - XXXXX		/89 - XXXXX		/90 - XXXXX	
	P/N	A DIA REF 3/ (inches)	P/N	A DIA REF 3/ (inches)	P/N	A DIA REF 3/ (inches)	P/N	A DIA REF 3/ (inches)
-10, -20	08W03	.270	09W03	.270	09W03	.270	09W03	.270
-10, -20	10W03	.332	11W03	.332	11W03	.332	11W03	.332
-10, -20	12W02	.332	13W02	.332	13W02	.332	13W02	.332
-10, -20	12W03	.458	13W03	.458	13W03	.458	13W03	.458
-10, -20	14W02	.458	15W02	.458	15W02	.458	15W02	.458
-11, -21	14W03	.582	15W03	.582	15W03	.582	15W03	.582
-11, -21	16W02	.520	17W02	.520	17W02	.520	17W02	.520
-11, -21	16W03	.645	17W03	.645	17W03	.645	17W03	.645
-11, -21	18W02	.645	19W02	.645	19W02	.645	19W02	.645
-12, -22	18W03	.770	19W03	.770	19W03	.770	19W03	.770
-11, -21	20W02	.645	21W02	.645	21W02	.645	21W02	.645
-12, -22	20W03	.832	21W03	.832	21W03	.832	21W03	.832
-12, -22	22W02	.704	23W02	.704	23W02	.704	23W02	.704
-13, -23	22W03	.958	23W03	.958	23W03	.958	23W03	.958
-12, -22	24W02	.770	25W02	.770	25W02	.770	25W02	.770
-13, -23	24W03	1.020	25W03	1.020	25W03	1.020	25W03	1.020

1/ -1X = Stainless steel spring band; -2X = Non-magnetic cobalt alloy spring band.
2/ See figure 2 – last five characters of the part number.
3/ See figure 3. Note that in addition to the A dimension DIA, the thickness of the braid shield must be considered. Shield braid thickness was accounted for in determining appropriate spring band size.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:15Z
Check the source to verify that this is the current version before use.

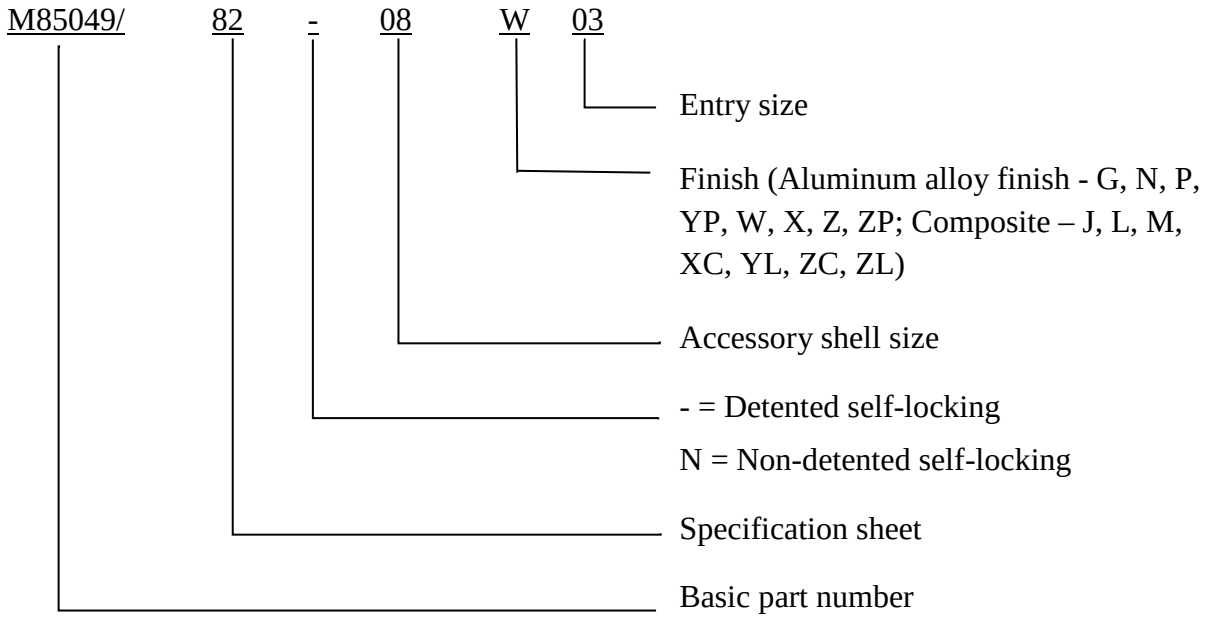


FIGURE 2. Part number example (M85049/82 through /90).

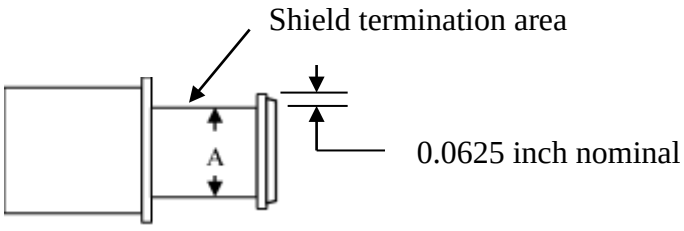


FIGURE 3. Typical M85049 backshell accessory.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:15Z
Check the source to verify that this is the current version before use.

3. Spring band application range. Table VII provides the spring band recommended minimum/maximum "A" dimension, outer diameter of the shield termination area for which the spring band may be used. Using that information, the compatible backshell can be selected from table VI.

TABLE VII. Spring band diameter range.

M32628/01 – XX 1/ Part number	A OD Range 2/ (Inches)	
	Min	Max
-10	.270	.458
-11	.520	.645
-12	.704	.832
-13	.895	1.145
-20	.270	.458
-21	.520	.645
-22	.704	.832
-23	.895	1.145

1/ -1X = Stainless steel spring band; -2X = Non-magnetic cobalt alloy spring band. See figure 4.

2/ OD range is for information only. Design and finish of the backshell shield termination configuration (finish, knurling, banding shoulder geometry and dimension, etc.) are also important for proper shield termination. Note that in addition to the OD dimension the thickness of the braid shield must be considered. Shield braid thickness was accounted for in determining appropriate spring band size.

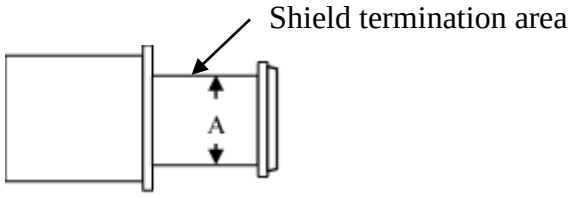


FIGURE 4. Shield termination/banding platform.

4. Spring installation guidance. See MIL-DTL-32628.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:15Z
Check the source to verify that this is the current version before use.

5. Patent notice.

U.S. PATENT NOTICE

The Government has a royalty-free license under the following patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

Patent No. U.S. 10,224,668 B2

Referenced documents. In addition to MIL-DTL-32628, this specification sheet references:

ASTM A666	SAE AMS5876	SAE AS85049/82
SAE AS85049/83	SAE AS85049/84	SAE AS85049/85
SAE AS85049/86	SAE AS85049/87	SAE AS85049/88
SAE AS85049/89	SAE AS85049/90	SAE AS85049/128

Custodians:

Army - CR
Navy - AS
Air Force - 85

Preparing activity:

Navy-AS

Project 5935-2019-015

Review activities:

Army - AV, MI
Navy - EC, OS, SH
Air Force - 11, 20
DLA - GS

Industry association:

SAE AE-8C1

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:15Z
Check the source to verify that this is the current version before use.